

JOB PORTAL MANAGEMENT SYSTEM

School of Computing

**Bachelor of Technology
in
Computer Science & Engineering**

By

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Full Stack Application Development

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May, 2026

BONAFIDE CERTIFICATE

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DECLARATION

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ABSTRACT

This project presents the design and development of a web-based Job Portal Application using Spring Boot, Thymeleaf, and MySQL. The system provides a platform for job seekers to search and apply for jobs, and for employers to post job openings and manage applications.

The application follows the Model-View-Controller (MVC) architecture, ensuring proper separation between the user interface, business logic, and data management. Spring Boot is used for backend development, Thymeleaf for dynamic web pages, and MySQL for database storage.

The system includes features such as user registration, login authentication, job posting, and application management. It improves the efficiency of recruitment processes and reduces manual effort.

Overall, the project demonstrates the effective use of modern web technologies in building a simple and efficient job portal system.

Keywords : Job Portal, Spring Boot, Thymeleaf, MySQL, MVC Architecture, Web Application

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LIST OF ACRONYMS AND ABBREVIATIONS

MVC	Model View Controller
API	Application Programming Interface
JPA	Java Persistence API
JDBC	Java Database Connectivity
SQL	Structured Query Language
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
HTTP	HyperText Transfer Protocol
DBMS	Database Management System

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Chapter 1

INTRODUCTION

1.1 Introduction

The rapid growth of the internet has significantly transformed the way recruitment and job searching are carried out. Traditional methods of job application and hiring are often time-consuming, inefficient, and lack proper organization. To overcome these challenges, a web-based Job Portal Application has been developed using modern technologies such as Spring Boot, Thymeleaf, and MySQL.

This application provides a centralized platform where job seekers can search and apply for jobs, while employers can post job vacancies and manage applications effectively. The system ensures seamless communication between users and improves the overall recruitment process. By automating various tasks, the application reduces manual effort and enhances efficiency.

The project follows the Model-View-Controller (MVC) architecture, which en-

ensures proper separation of concerns between different layers of the application. The frontend is designed using Thymeleaf templates, the backend logic is implemented using Spring Boot, and MySQL is used for storing and managing data.

1.2 Aim of the Project

The main aim of this project is to design and develop a user-friendly and efficient Job Portal system that simplifies the recruitment process for both job seekers and employers.

The objectives of the project include:

- To provide a platform for users to register and log in securely.
- To enable job seekers to search and apply for jobs.
- To allow employers to post job vacancies and manage applicants.
- To maintain user and job data efficiently using a database system.
- To implement a scalable and maintainable web application using Spring Boot.

1.3 Scope of the Project

The scope of this project includes the development of a web-based application that facilitates job searching and recruitment activities. The system is designed to handle multiple users simultaneously and provide a responsive and interactive user experience.

The application covers:

- User registration and authentication.
- Job posting and management by employers.

- Job search and application by job seekers.
- Database management using MySQL.
- Basic security features for user data protection.

However, the system currently focuses on core functionalities and can be extended in the future with advanced features such as email notifications, resume uploads, and AI-based job recommendations.

1.4 Framework

The project is developed using the Spring Boot framework, which simplifies the development of Java-based web applications. It follows the MVC (Model-View-Controller) architecture:

- **Model:** Represents the data layer, including entities and database interactions.
- **View:** Developed using Thymeleaf templates to provide a dynamic user interface.
- **Controller:** Handles user requests and controls the application flow.

Spring Boot provides features such as dependency injection, embedded servers, and easy integration with databases, making it suitable for building scalable and efficient applications.

The overall system architecture of the Job Portal Application is shown in [Figure1.1]. It illustrates the interaction between the user interface, controller, and database layers, ensuring efficient data flow within the system.

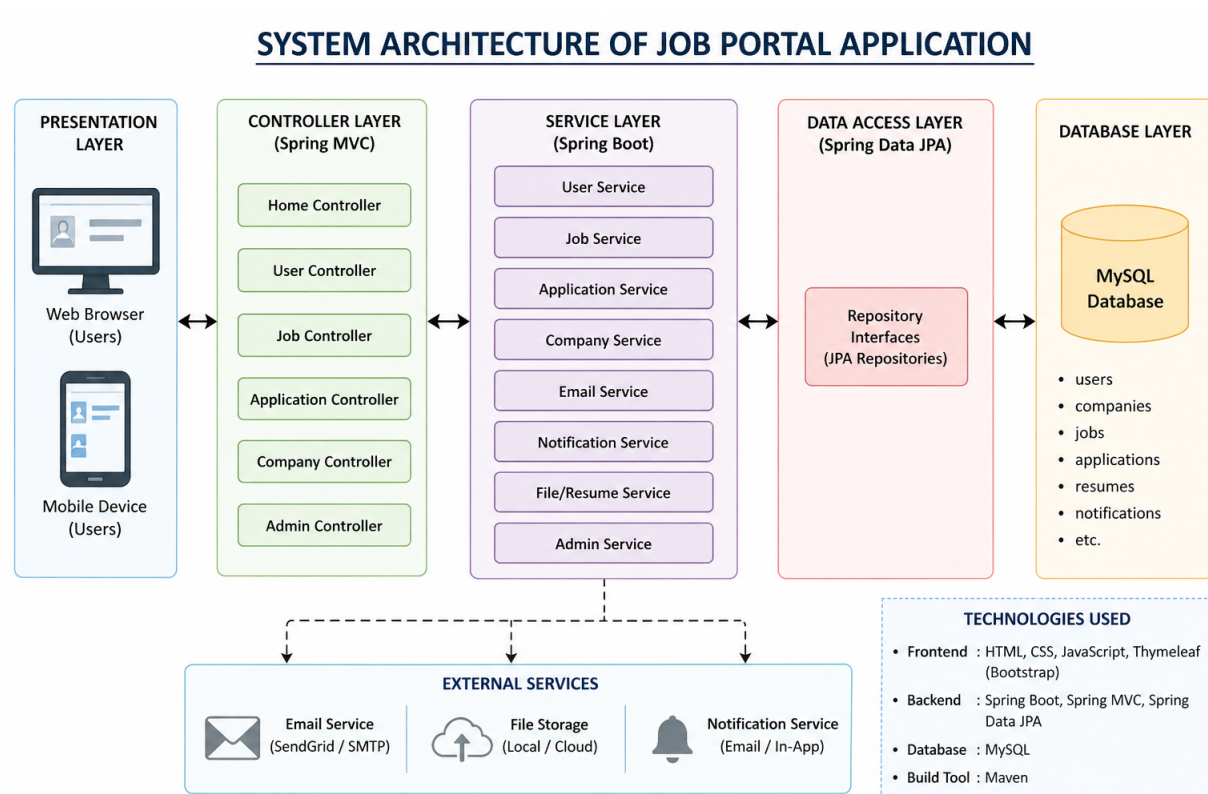


Figure 1.1: System Architecture of Job Portal Application

Chapter 2

SURVEY OF SIMILAR APPLICATION IN MARKET

The rapid advancement of web technologies has led to the development of various online job portal platforms that connect job seekers with employers. These platforms provide efficient solutions for recruitment and job searching. A study of existing systems helps in understanding their features, advantages, and limitations, which in turn assists in designing an improved application.

Some of the popular job portal applications available in the market are:

2.1 LinkedIn

LinkedIn is one of the most widely used professional networking platforms. It allows users to create professional profiles, connect with industry experts, and apply for jobs. Recruiters can post job openings and search for candidates based on their

skills and experience.

Advantages:

- Strong professional networking features
- Job recommendations based on user profiles
- Easy communication between recruiters and candidates

Limitations:

- Complex interface for new users
- Many features require premium subscription

2.2 Naukri

Naukri is a popular job portal widely used in India. It provides a platform for job seekers to upload resumes and apply for jobs, while employers can search for suitable candidates.

Advantages:

- Large database of job listings
- Resume upload and profile management
- Simple job search functionality

Limitations:

- Limited personalization of job recommendations
- Frequent notifications may be distracting

2.3 Indeed

Indeed is a global job search platform that aggregates job listings from various sources. It allows users to search for jobs based on location, salary, and job type.

Advantages:

- Wide range of job listings from multiple sources
- Easy-to-use interface
- Free access for job seekers

Limitations:

- Duplicate job postings may appear
- Limited direct interaction with employers

2.4 Comparison with Proposed System

The proposed Job Portal Application aims to overcome some of the limitations of existing systems by providing a simple, user-friendly interface and essential functionalities without unnecessary complexity. Unlike existing platforms, the system focuses on ease of use, efficient job management, and streamlined communication between users.

Key Improvements:

- Simplified user interface
- Focused features for academic-level implementation
- Efficient database management using MySQL
- Secure authentication using Spring Boot

Thus, the proposed system provides a balanced solution by combining essential features of existing applications while maintaining simplicity and efficiency.

Chapter 3

FRAMEWORK DESCRIPTION

3.1 Frontend Implementation

3.1.1 Environment Setup

The frontend of the application is developed using HTML, CSS, and Thymeleaf templates. Thymeleaf is a server-side Java template engine used for rendering dynamic web pages in Spring Boot applications.

The development environment includes:

- Visual Studio Code (VS Code) as the code editor
- Web browser (Chrome/Edge) for testing
- Maven for project dependency management

Thymeleaf integrates seamlessly with Spring Boot, enabling dynamic data rendering from the backend to the frontend.

3.1.2 Structure of the Project

The frontend files are organized within the project as follows:

- **templates/** – Contains Thymeleaf HTML files such as login, register, dashboard, and profile pages.
- **static/** – Contains static resources like CSS, JavaScript, and images.

Each HTML page is designed to interact with backend controllers and display dynamic content to users.

3.1.3 Execution Procedure

- Start the Spring Boot application.
- Open a web browser and navigate to the application URL (e.g., <http://localhost:8081>).
- The frontend pages are loaded dynamically using Thymeleaf.
- Users can interact with forms such as login, registration, and job applications.

3.2 Backend Implementation

3.2.1 Environment Setup

The backend is developed using Spring Boot, a Java-based framework that simplifies the creation of web applications.

The environment setup includes:

- Java Development Kit (JDK 17 or above)
- Spring Boot framework
- Maven for dependency management
- MySQL Driver (mysql-connector)

Spring Boot provides built-in support for REST APIs, dependency injection, and database connectivity.

3.2.2 Structure of the Project

The backend follows the MVC (Model-View-Controller) architecture and is organized into the following packages:

- **controller/** – Handles user requests and responses.
- **service/** – Contains business logic of the application.
- **repository/** – Handles database operations using JPA.
- **security/** – Manages authentication and authorization.

3.2.3 Execution Procedure

- Run the application using Maven command: `mvn spring-boot:run`.
- Spring Boot initializes the embedded server.
- Controllers handle incoming HTTP requests.
- Responses are sent back to the frontend.

The working flow of the system is illustrated in [Figure3.1]. It shows how requests move from the frontend to the backend and how responses are returned after processing.

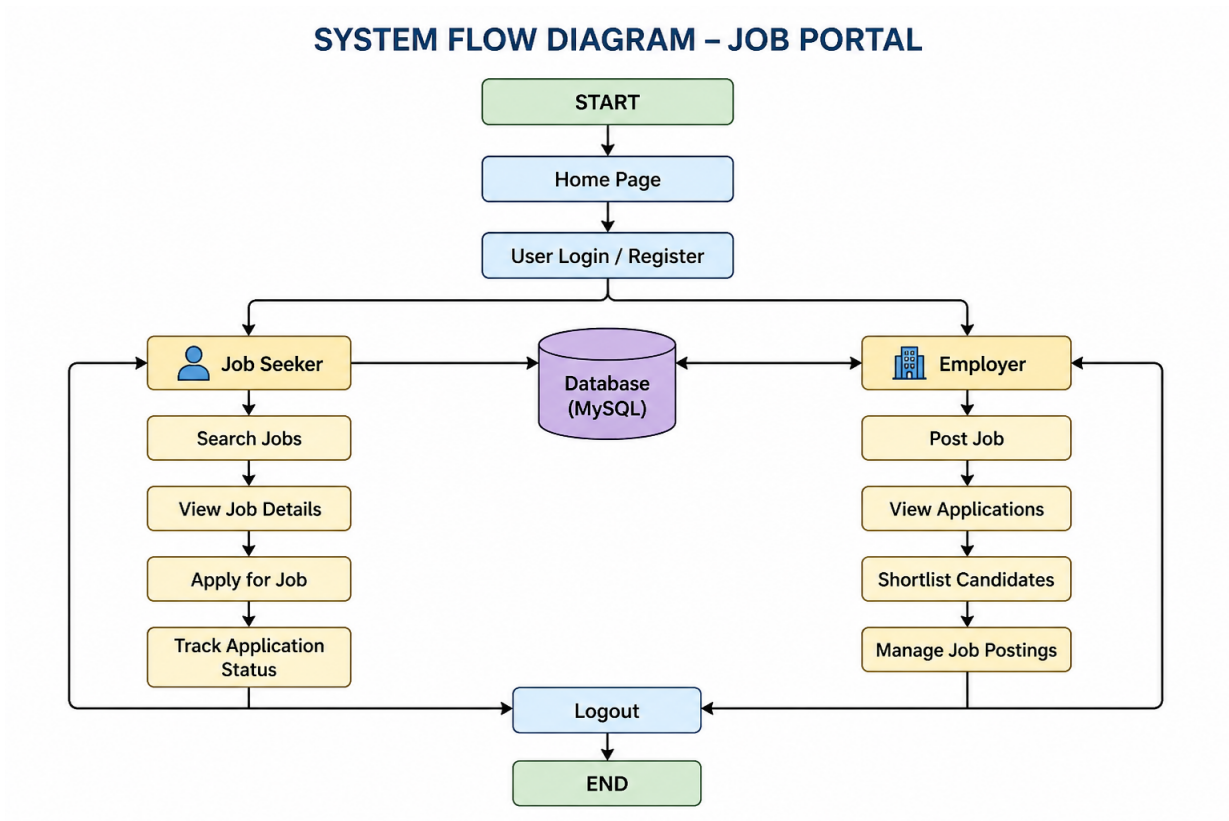


Figure 3.1: System Flow Diagram

3.3 Database Implementation

3.3.1 Database Configuration

The database used in this project is MySQL. It is configured using the application.properties file in Spring Boot.

The configuration includes:

- Database URL (jdbc:mysql://localhost:3306/student_db)
- Username and password
- Hibernate properties for automatic table creation and updates

Spring Boot connects to MySQL using JDBC and manages data using JPA (Java Persistence API).

3.3.2 Schema Structure

The database schema is designed to store user and job-related information efficiently. It includes multiple tables that represent different entities in the system.

3.3.3 Tables created

The main tables created in the database include:

- **users** – Stores user details such as ID, name, email, and password.
- **jobs** – Stores job postings including title, description, and company details.
- **applications** – Stores job application details linking users and jobs.

Chapter 4

METHODOLOGIES

4.1 Project Architecture

The project follows the Model-View-Controller (MVC) architecture, which separates the application into three main components: Model, View, and Controller. This separation improves maintainability, scalability, and code organization.

- **Model:** Represents the data layer, including entity classes and database interactions using JPA.
- **View:** Implemented using Thymeleaf templates to display dynamic content to users.
- **Controller:** Handles incoming requests, processes them using services, and returns appropriate responses.

The architecture ensures a smooth flow of data between the user interface and

the database. The overall structure of the system is shown in [Figure4.1], which illustrates the interaction between different layers of the application.

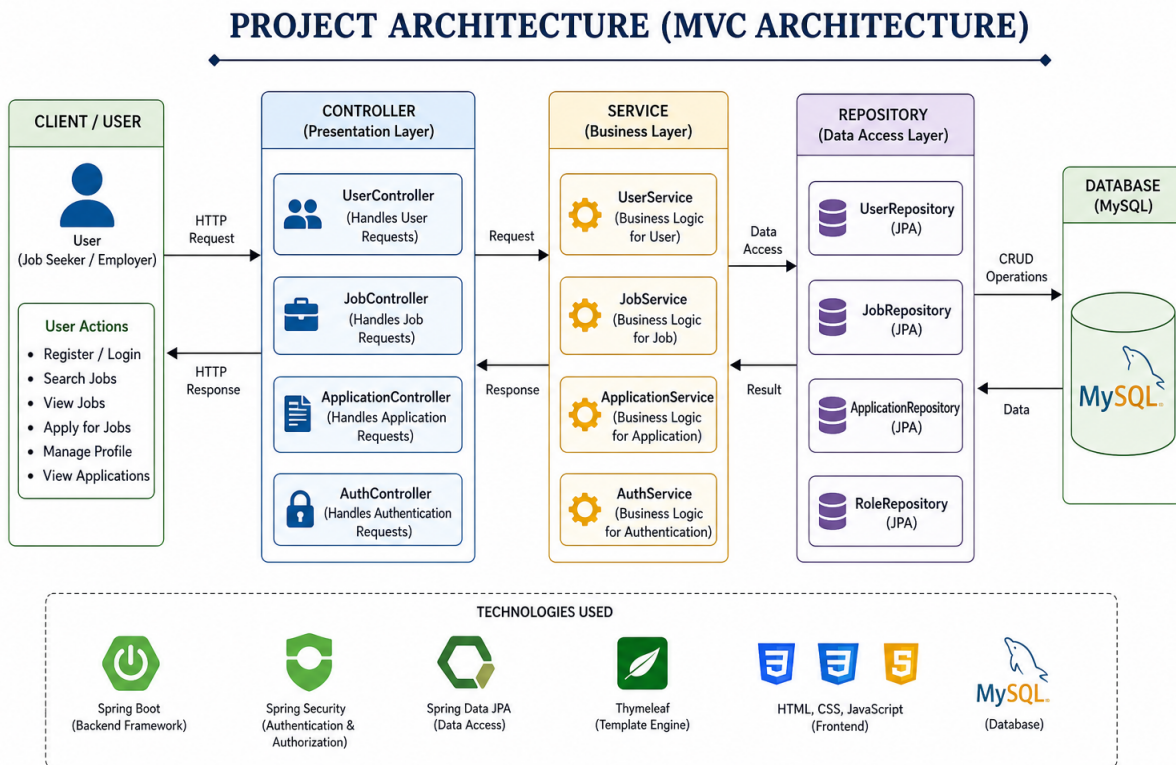


Figure 4.1: MVC Architecture of Job Portal System

4.2 DataFlow Diagram

The Data Flow Diagram (DFD) represents how data moves through the system. It shows the interaction between users, the application, and the database.

- Users provide input through the frontend (login, registration, job search).
- The controller processes the request and sends it to the service layer.
- The service layer interacts with the repository to access the database.
- The database returns the required data.
- The result is displayed to the user through the frontend.

The data movement within the system is illustrated in [Figure4.2], showing how information flows between different components during execution.

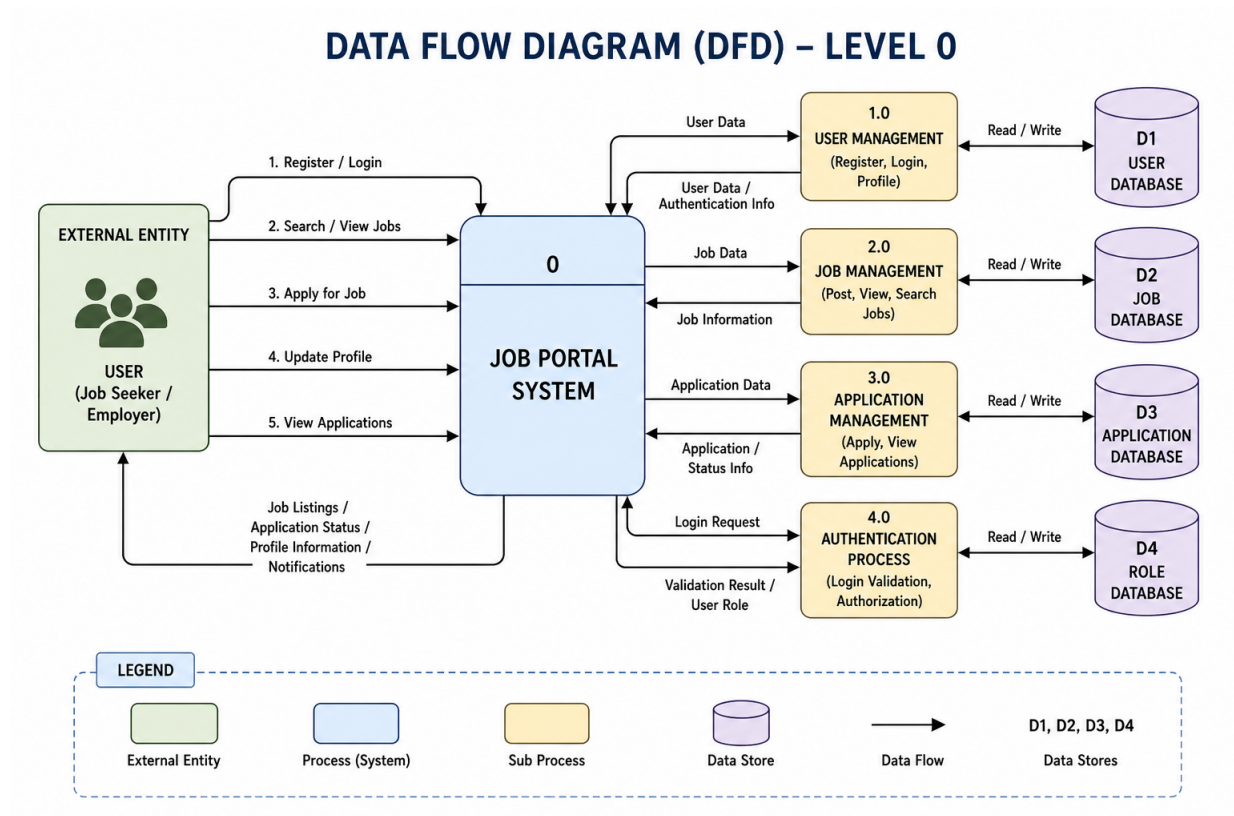


Figure 4.2: Data Flow Diagram

4.3 Module 1: User Management

This module handles user-related functionalities such as registration, login, and profile management.

- User registration with basic details
- Secure login authentication
- Profile viewing and updating

4.4 Module 2: Job Management

This module is responsible for managing job postings.

- Employers can post job vacancies
- Job details such as title, description, and company information are stored
- Users can browse available jobs

4.5 Module 3: Application Management

This module allows users to apply for jobs and manage applications.

- Users can apply for jobs
- Applications are stored in the database
- Employers can view applicants

Advantages of this Project

The proposed system provides several advantages:

a) User-Friendly Interface: The application provides a simple and easy-to-use interface for both job seekers and employers.

b) Efficient Data Management: All data is stored and managed using MySQL, ensuring quick access and reliability.

c) Scalable Architecture: The use of Spring Boot and MVC architecture allows the system to be easily extended with additional features.

Disadvantages

- Limited advanced features such as AI-based job recommendations.
- No email notification system for job updates.
- Requires internet connection for access.

Chapter 5

RESULTS AND DISCUSSIONS

The developed Job Portal Application was successfully implemented using Spring Boot, Thymeleaf, and MySQL. The system was tested for various functionalities including user registration, login, job posting, and job application[table 5.1].

The application provides a smooth and user-friendly interface, allowing users to easily interact with the system. The backend processes requests efficiently, and the database ensures secure and reliable data storage.

Observed Results:

- Users are able to register and log in successfully.
- Employers can post job vacancies without errors.
- Job seekers can search and apply for jobs.
- Data is stored and retrieved correctly from MySQL database.
- The system performs efficiently with minimal response time.

Table 5.1: Comparison of Existing Systems and Proposed System

Features	Existing Systems	Proposed System
User Interface	Complex	Simple and User-Friendly
Cost	Paid Features	Free to Use
Job Search	Available	Available
Job Application	Available	Available
Customization	Limited	Flexible
Performance	Moderate	Efficient

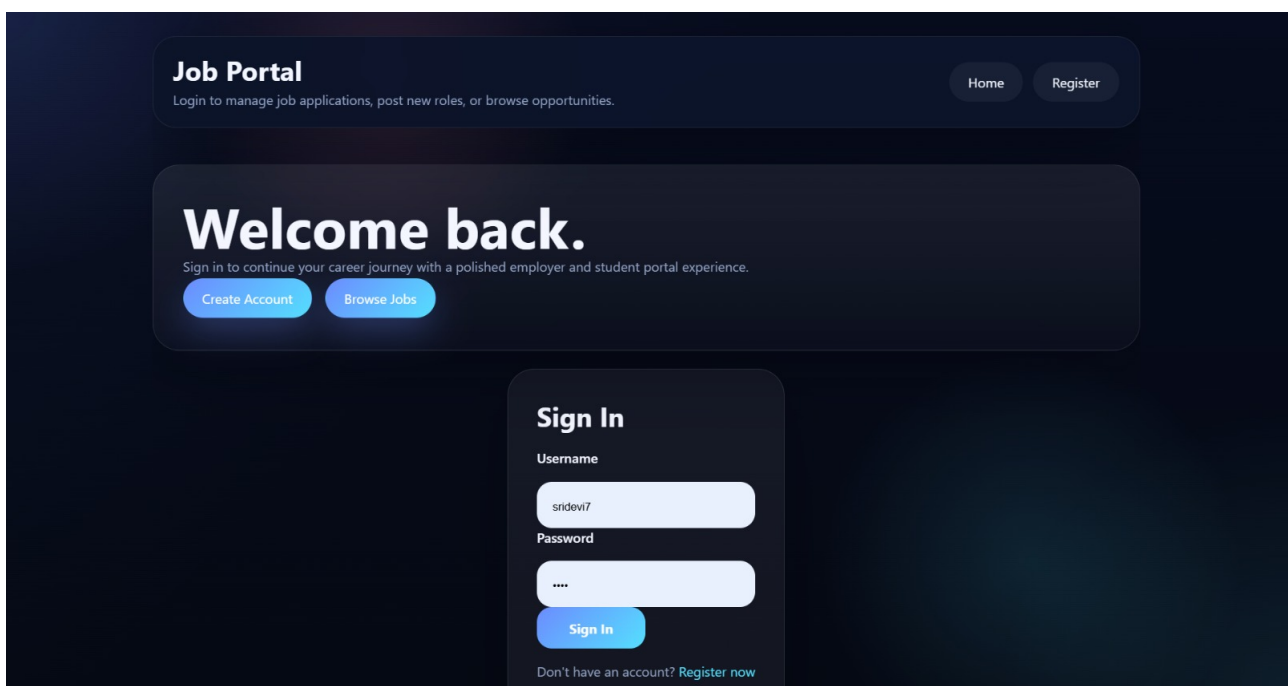


Figure 5.1: Login Output Screen

Discussion:

The results indicate that the proposed Job Portal Application performs efficiently and meets the intended objectives. Compared to existing systems, it provides a simplified interface and focuses on essential functionalities without unnecessary complexity.

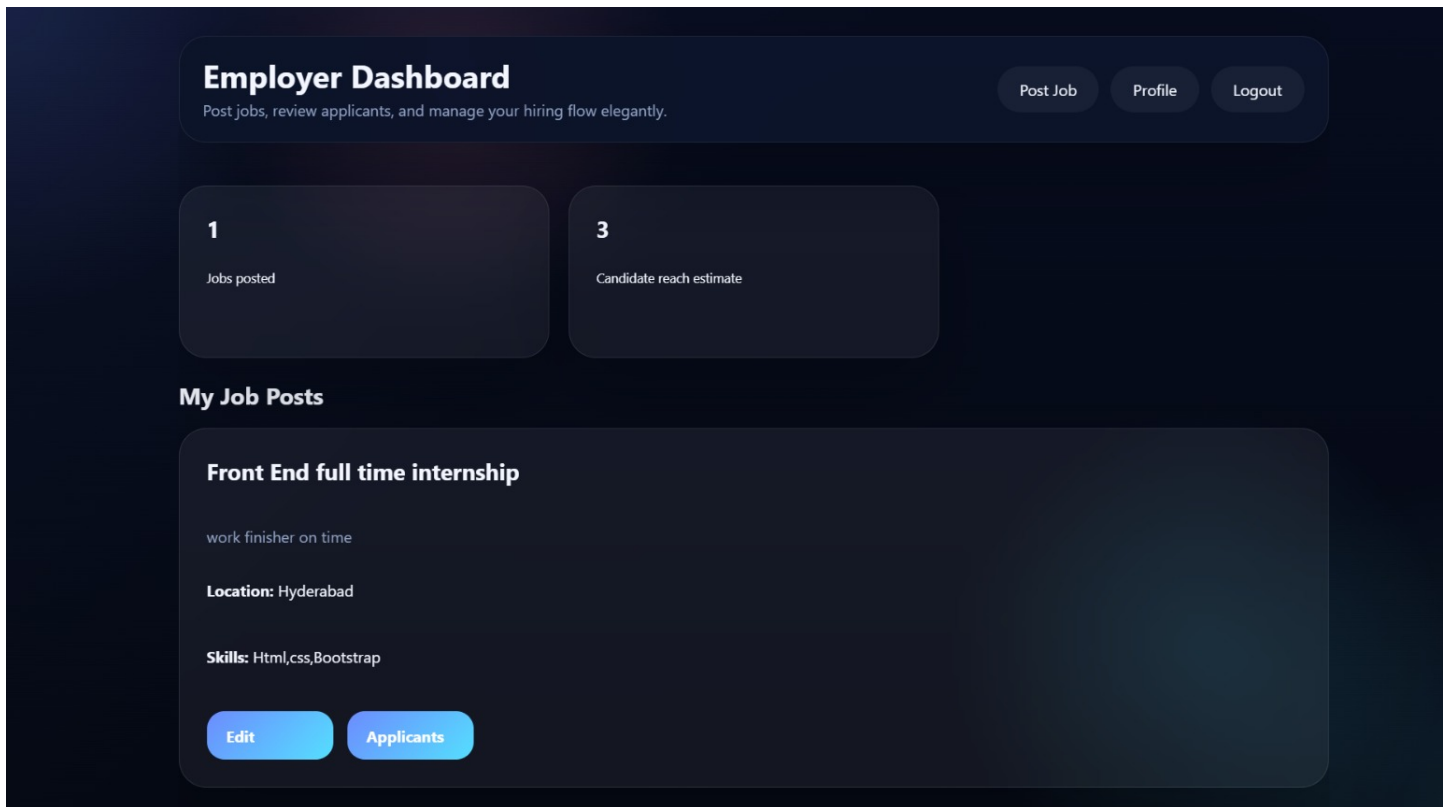


Figure 5.2: Employer Dashboard Output Screen

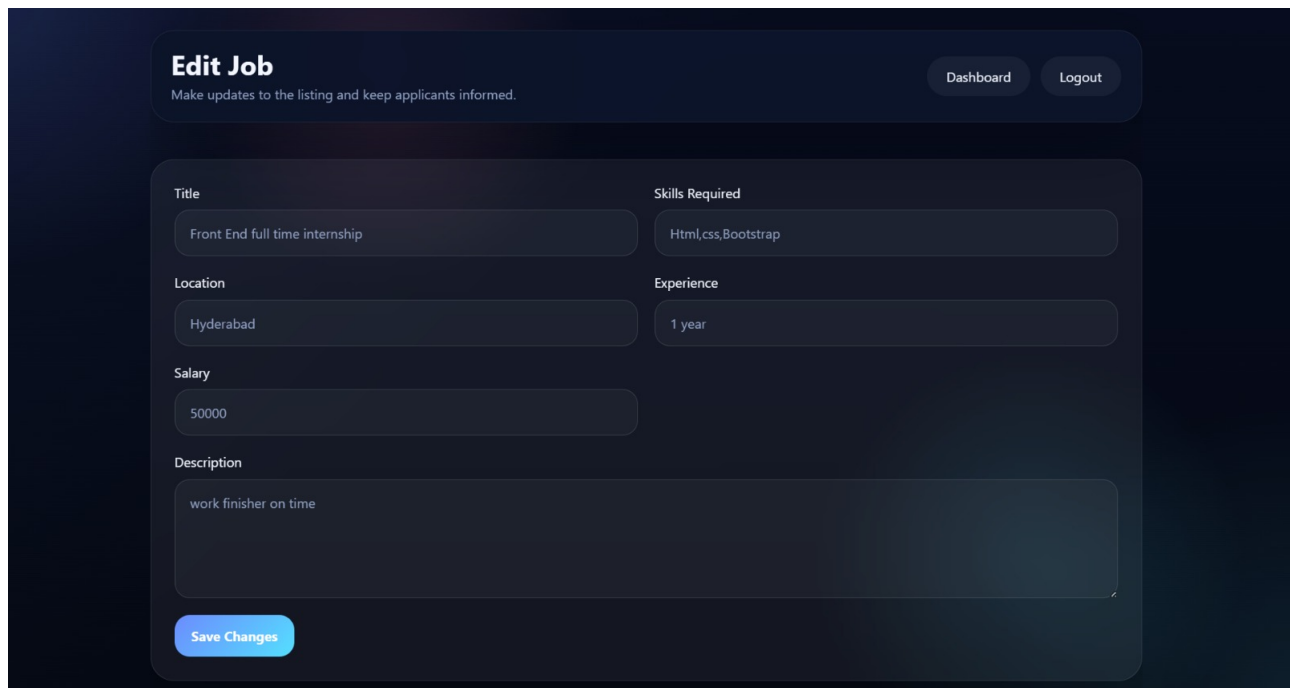


Figure 5.3: Edit job Output Screen

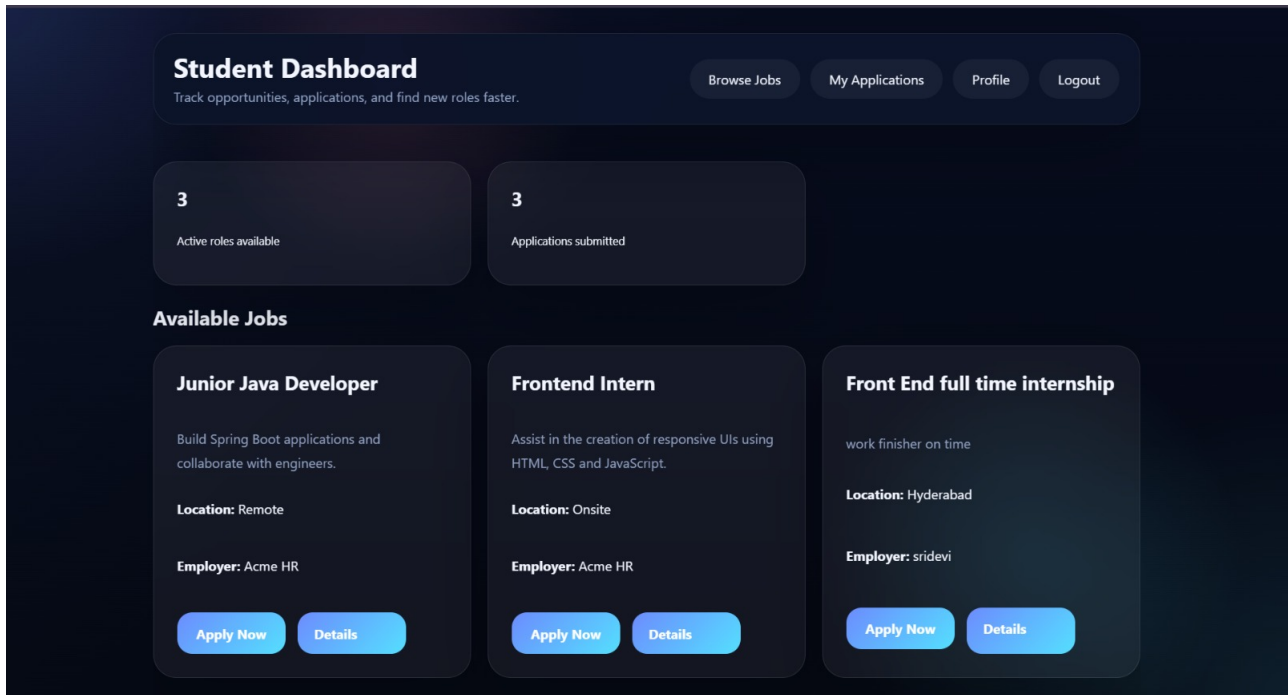


Figure 5.4: Student Dashboard Output Screen

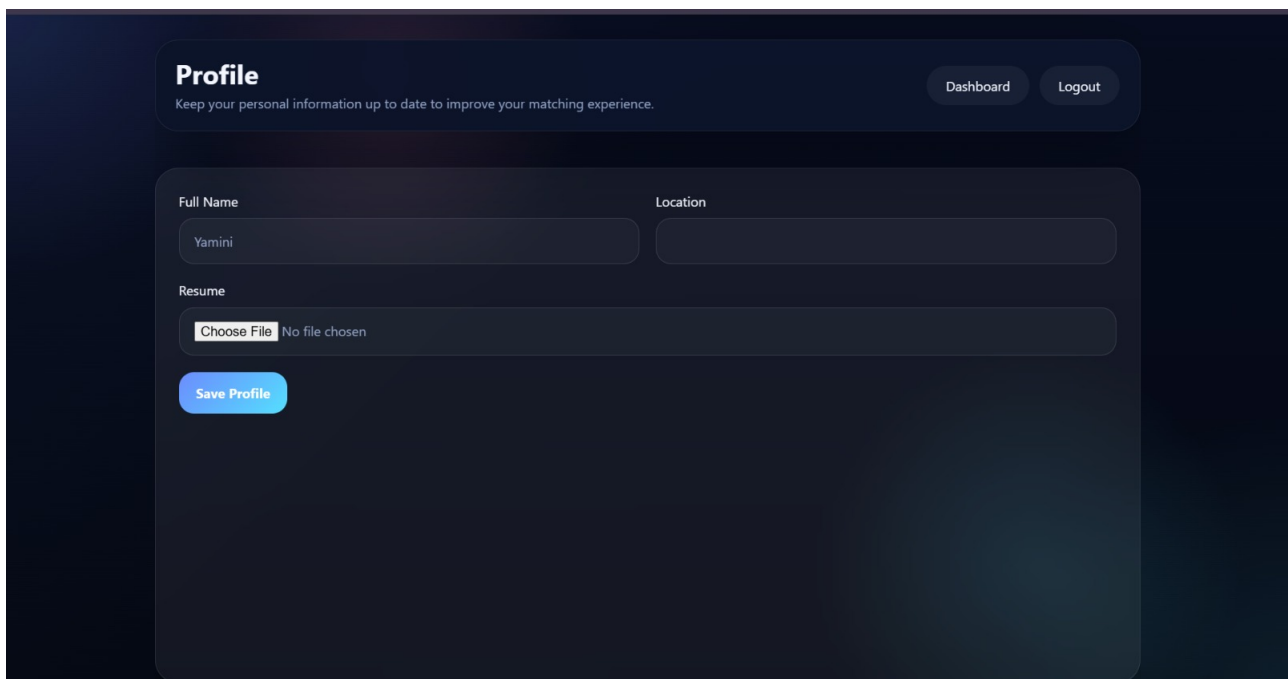


Figure 5.5: Student profile Output Screen

Chapter 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Conclusion

The Job Portal Application was successfully designed and implemented using Spring Boot, Thymeleaf, and MySQL. The system provides an efficient platform for job seekers and employers to interact, reducing the complexity of traditional recruitment processes.

The application supports essential functionalities such as user registration, login authentication, job posting, and job application. The use of MVC architecture ensures a well-structured and maintainable system. The integration of frontend and backend technologies enables smooth data flow and dynamic content rendering.

Overall, the project meets its objectives by providing a simple, scalable, and user-friendly solution. It demonstrates the effective use of modern web technologies in

building real-world applications.

6.2 Future Enhancements

Although the system performs well, several improvements can be made in the future to enhance its functionality:

- **Email Notifications:** Notify users about job updates and application status.
- **Resume Upload Feature:** Allow users to upload and manage resumes.
- **Advanced Search Filters:** Improve job search with filters like salary, location, and experience.
- **Admin Dashboard:** Add admin control for managing users and job postings.
- **Mobile Application:** Develop an Android/iOS version of the system.
- **AI-based Recommendations:** Suggest jobs based on user skills and preferences.

Chapter 7

ADDRESSING COMPLEX ENGINEERING PROBLEM AND SDG

7.1 Mapping of the Project to Complex Engineering Problem (CEP)

Level	Attribute	Characteristics	WKs	Justification
WP1	Depth of Knowledge	In-depth knowledge	WK3,4,5,6	Requires Java, Spring Boot, JPA, and database design for Job Portal system.
WP2	Conflicting Requirements	Tech & non-tech conflicts	WK4,5,6	Balances security with usability for job seekers and recruiters.
WP3	Depth of Analysis	Analytical design	WK3,4,5	Includes ER design, APIs, resume upload, and application logic.
WP4	Familiarity	Partially new	WK4,8	Job matching and tracking beyond CRUD systems.

7.2 Mapping of the Project to Complex Engineering Activities(CEA)

EA No.	Attribute	Characteristics	Justification
EA1	Resources	Diverse tech	Uses Java, Spring Boot, MySQL, HTML, CSS, JS.
EA2	Interaction	Complex modules	Handles jobs, applications, login, and uploads.
EA3	Innovation	Modern tech	Role-based dashboards, filtering, email alerts.
EA4	Society Impact	User benefit	Improves hiring and job accessibility.
EA5	Familiarity	Beyond basics	Combines tracking, resume, and admin system.

7.3 SDG Mapping for the Project

SDG No.	SDG Title	Relevance
SDG 8	Decent Work	Connects job seekers and employers for employment.
SDG 9	Innovation	Uses modern web technologies for recruitment.
SDG 10	Reduced Inequality	Equal job access for all users.
SDG 16	Strong Institutions	Ensures security using authentication system.

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7.4 SOURCE CODE IMPLEMENTATION

7.4.1 Login Page

```
<!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org">
<head>
  <meta charset="UTF-8" />
  <title>Login - Job Portal</title>
  <link rel="stylesheet" th:href="@{/css/style.css}" />
</head>
<body>
<div class="container page-grid">
  <header>
    <div>
      <h1>Job Portal</h1>
      <p>Login to manage job applications, post roles, or browse jobs.</p>
    </div>
    <nav>
      <a th:href="@{/}">Home</a>
      <a th:href="@{/register}">Register</a>
    </nav>
  </header>

  <section class="hero">
    <div>
      <h1>Welcome back</h1>
      <p>Sign in to continue your journey.</p>
    </div>
  </section>

  <section class="auth-form">
    <h1>Sign In</h1>
    <form method="post" th:action="@{/login}">
      <label>Username</label>
      <input type="text" name="username" required />

      <label>Password</label>
      <input type="password" name="password" required />

      <button type="submit">Sign In</button>
    </form>

    <p>Don't have an account? <a th:href="@{/register}">Register</a></p>

    <div th:if="${param.error}">Invalid username or password</div>
    <div th:if="${param.logout}">Logged out successfully</div>
    <div th:if="${param.registered}">Registration successful</div>
  </section>
</div>
</body>
</html>
```

7.4.2 Student Dashboard

```
<!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org">
<head>
  <meta charset="UTF-8" />
  <title>Student Dashboard</title>
  <link rel="stylesheet" th:href="@{/css/style.css}" />
</head>
<body>
<div class="container page-grid">
  <header>
    <h1>Student Dashboard</h1>
    <nav>
      <a th:href="@{/}">Browse Jobs</a>
      <a th:href="@{/applications}">My Applications</a>
      <a th:href="@{/profile}">Profile</a>
      <a th:href="@{/logout}">Logout</a>
    </nav>
  </header>

  <section>
    <h2>Available Jobs</h2>
  </section>
</div>
```



```

        <div th:each="job : ${jobs}">
            <h3 th:text="${job.title}"></h3>
            <p th:text="${job.description}"></p>
            <p><strong>Location:</strong> <span th:text="${job.location}"></span></p>
            <p><strong>Employer:</strong> <span th:text="${job.employer.fullName}"></span></p>

            <form th:action="@{'/jobs/' + ${job.id} + '/apply'}" method="post">
                <button type="submit">Apply Now</button>
            </form>
        </div>
    </section>
</div>
</body>
</html>

```

7.4.3 Application Tracking Page

```

<!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org">
<head>
    <meta charset="UTF-8" />
    <title>My Applications</title>
    <link rel="stylesheet" th:href="@{/css/style.css}" />
</head>
<body>
<div class="container page-grid">
    <header>
        <h1>My Applications</h1>
        <nav>
            <a th:href="@{/dashboard}">Dashboard</a>
            <a th:href="@{/profile}">Profile</a>
            <a th:href="@{/logout}">Logout</a>
        </nav>
    </header>

    <section>
        <div th:if="${applications.isEmpty()}">
            No applications yet.
        </div>

        <div th:each="app : ${applications}">
            <h3 th:text="${app.job.title}"></h3>
            <p><strong>Employer:</strong> <span th:text="${app.job.employer.fullName}"></span></p>
            <p><strong>Status:</strong> <span th:text="${app.status}"></span></p>
            <p><strong>Applied On:</strong> <span th:text="${app.appliedAt}"></span></p>
        </div>
    </section>
</div>
</body>
</html>

```